



Cansat 2026

Preliminary Design Review (PDR)

Team Mechaholics
Team ID: 2026-INSPACE-CANSAT-006



Acronyms

Acronyms	Definition
3D	Three Dimensional
CDH	Communication & Data Handling
CSV	Comma Separated Value
dB	Decibel
dBi	Decibel Isotropic
DCS	Descent Control Subsystem
EPS	Electrical Power Subsystem
FSW	Flight Software
G	Gravitational Force Constant
g	Gravitational force

Acronyms	Definition
GPS	Global Positioning System
IMU	Internal Measurement Unit
Li-Ion	Lithium-Ion
MCU	Microcontroller Unit
m/s	Meters/Second
PLA+	Polylactic Acid +
RAM	Random Access Memory
RPM	Revolutions Per Minute
RTC	Real Time Clock



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7.1 TEAM COMPOSITION AND MANAGEMENT



TEAM COMPOSITION AND MANAGEMENT

Team Member details	Branch	Year of Graduation	Role in the team
Prof.Anjaneyulu	ECE	N/A	Faculty Advisor
Jacob Sadeesh	ECE	2028	Team Lead
Ankathi Hari Sreenu Varma	ECE VLSI	2027	Team Member
Rudransh Singh	Mechanical	2027	Team Member
Don Roy Chacko	CSE	2028	Team Member



TEAM COMPOSITION AND MANAGEMENT

Team Member Details	Branch	Year of Graduation	Role in the team
Jaikishan Gopalakrishnan	ECE	2027	Team Member
Sriniklesh Guthula	ECE	2029	Team Member
Nandakishore Varma Buddharaju	ECE	2029	Team Member
Gayathri Amarkumar	EEE	2029	Team member



7.2 Mission Overview



Mission objectives

1.BASIC REQUIREMENTS

- **Mass:** ≤ 1.0 kg (± 0.05 kg)
- **Dimensions:** Must fit within **0.15m × 0.4 m** cylindrical envelope
- **Structure:** Any material/design permitted; no sharp edges
- **Visibility:** Fluorescent **pink/red/orange** body with **Indian flag** markings
- **Independence:** Must not rely on the rocket airframe for operation
- **Made in India:** Indigenous components receive additional weightage
- **NavIC Support:** GNSS with **NavIC** receives additional weightage
- **Mandatory Sensors:** Position, altitude, pressure, temperature, orientation, power, and system status
- **Telemetry:** Near real-time data display on ground station UI
- **Data Logging:** Onboard SD card backup during telemetry loss



Mission Guidelines

2. STRUCTURE REQUIREMENTS

- Enclosed electronics with structural housing
- Survive **15 G launch** and **30 G shock**
- Secure PCB mounting
- Multilingual identification markings
- Audio beacon (≥ 92 dB) for recovery

3. POWER SYSTEM OBJECTIVES

- External ON/OFF switch with status indication
- Minimum 2-hour launch-pad endurance
- Approved battery chemistry compliance
- Quick-access battery compartment
- Shock-resistant battery connections



Mission Guidelines

4. DESCENT & RECOVERY REQUIREMENTS

- **Dual Descent System:** Two-stage descent control mechanism required
- **Primary Recovery:** Immediate parachute deployment after ejection
- **Secondary Recovery:** Team-selected mechanism (streamer, parachute, glider, rotor, etc.)
- **Descent Rates:** Primary ≤ 20 m/s; secondary 1–3 m/s
- **Secondary Activation:** Deploy at **600 m $\pm$ 10 m** altitude
- **Parachute Integration:** Primary parachute attached directly to CAN-7USAT body; no ejection system
- **Gyroscopic Stabilization:** Mechanical gyroscopic wheels required during descent
- **Safety Compliance:** No pyrotechnics or explosive devices; green propulsion permitted
- **Structural Robustness:** Descent system must survive **30 G shock loads**
- **Protected Electronics:** All electronics enclosed; only sensors may be exposed



System Overview



CAN-7USAT configuration

The CAN-7USAT is a compact cylindrical satellite that integrates avionics, communication, imaging, data logging, power management, and recovery subsystems within a single self-contained structure. The primary mission of the CAN-7USAT is to acquire and transmit real-time flight data while recording mission parameters for post-flight analysis.

The onboard electronics are centered around the STM32F405 flight controller, which interfaces with the BMP585 barometric sensor, BNO086 inertial measurement unit, NEO-M9N GNSS receiver, LoRa communication module, MicroSD data logger, and M5StackCAM imaging payload. These components are arranged within the structure to maximize space utilization while maintaining accessibility, structural stability, and reliable operation throughout the mission.

The recovery system employs a two-stage descent strategy consisting of a primary parachute deployed immediately after ejection and a secondary parachute deployed at approximately 600 m altitude to further reduce the descent rate. This configuration enables stable flight, continuous data collection, and safe landing of the CAN-7USAT.



Major Components(1/2)

Component	Category
STM32F405	Flight computer
BMP585	Altimeter
BNO086	IMU
NEO-M9N	GNSS
SX1276 LoRa	Telemetry radio
M5StackCAM(OV2640)	Camera
microSD (SPI)	Flight computer



Major Components(2/2)

STM32F405:

Main flight controller responsible for sensor acquisition, telemetry handling, mission control, and data logging.

BMP585:

High-precision barometric sensor for altitude and temperature measurements.

BNO086:

9-axis IMU providing orientation, acceleration, angular velocity, and magnetic field measurements.

NEO-M9N:

Multi-constellation GNSS receiver providing position, altitude, velocity, and timing information.

SX1276 LoRa:

Long-range wireless communication module operating at 868 MHz for telemetry transmission.

M5StackCAM (OV2640):

Captures images and video during flight operations.

MicroSD Card (SPI):

Stores sensor and telemetry data for post-flight analysis.



The mechanical structure of the CAN-7USAT has been designed such that it would meet all dimensional requirements and weight restrictions while providing structural strength throughout the processes of launching, deployment, descent, and retrieval. The cylinder shape allows efficient packaging of all the different components of the satellite. The mounting structures inside have been provided in such a way that all parts of the satellite remain in their place while in flight.

The mechanical drawing shows the dimensions of the CAN-7USAT, its structural shape, and how all the parts will be arranged. A lot of attention has been paid to ensuring proper balance of the center of gravity, using the available space efficiently, and making sure that it is easy to assemble.

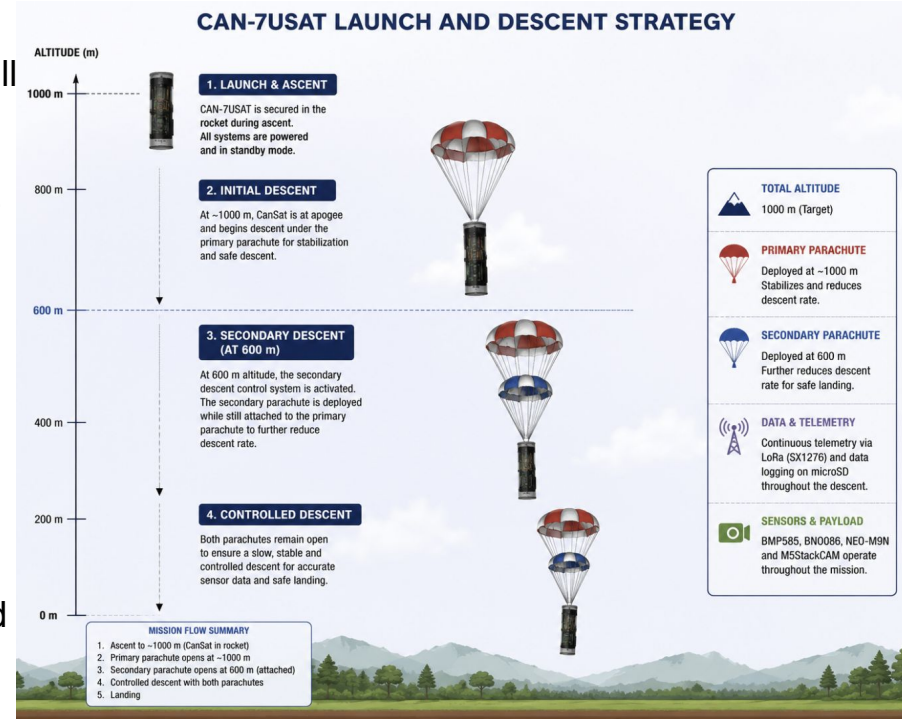


Launch and Descent Strategy(1/2)

- The integration process will be carried out by installing the satellite inside the payload bay of the launch vehicle where it will be kept in secure conditions throughout the launch process. In addition, all onboard systems will be switched on and ready for use and withstand the conditions in which it is launched into space.
- After reaching the required altitude and being ejected from the launch vehicle, the CAN-7USAT starts to execute the program sequence in order to successfully conduct its mission. Right after that, the main parachute will deploy and help the satellite to slow down its speed of descent and provide a relatively safe operating condition. During the descent, the STM32F405 flight controller will collect and transmit telemetry data from the BMP585 barometric sensor, BNO086 IMU, and NEO-M9N GNSS receiver. At the same time, telemetry data will be uploaded to the onboard microSD card for further usage.

Launch and Descent Strategy(2/2)

- When the altitude decreases to about 600 m above the ground, the secondary descent control system will become active and the secondary parachute will deploy, thus, further decreasing the speed of the descent. Such a descent control system will help not only provide more stable flight but also improve telemetry and imagery collection as well.
- Throughout the entire process of the descent, the onboard M5StackCAM imaging system will record images and video, which can be used to achieve the objectives of the satellite. Thus, the described descent process of the satellite will guarantee the safety of both onboard systems and equipment and allow the completion of the mission.





Post-launch Recovery

- After landing, the CAN-7USAT payload will move into the recovery stage. The built-in NEO-M9N GNSS receiver will constantly track the coordinates, with the last position sent from the CAN-7USAT to the ground station through the LoRa telemetry chain. By comparing these coordinates, it will be possible to estimate the landing site and locate the spacecraft on-site.
- In order to help localize the spacecraft during post-landing recovery, an audio beacon will be included in the system. The beacon will be emitting constant audio tones, which the ground crew will be able to detect by directional sound localization, even if the payload cannot be visually identified due to challenging environmental conditions, terrain, or foliage.
- The implementation of a reliable GPS tracker paired with the use of an audio beacon for payload localization will ensure a successful landing and recovery operation in most scenarios. After recovering the device, the microSD card stored inside will be removed from the payload for further analysis.



Data Retrieving and Analysis

- Once the CAN-7USAT is recovered successfully, all data on board the spacecraft will be retrieved for post-recovery evaluation purposes. The main data source would include the microSD card which holds high resolution telemetric data such as the GNSS positions, altitudes, and timestamps as well as other sensors' data and dynamics of descent during the mission.
- All data retrieved will be analyzed to determine how well the entire system performed, whether any design assumptions held and ways in which the system could be improved further.



7.3 SUBSYSTEM DETAILS



I. Payload Subsystem



Payload Subsystem

- Payload subsystem of the CAN-7USAT is responsible for data acquisition, processing, storage and transmission during entire course of flight mission. The main goal of the payload is the acquisition of altitude, temperature, orientation and motion data during descent. Another payload goal is to obtain aerial photos and video using camera.
- BMP585 barometer, which will be used to detect altitude and temperature, BNO086 inertial measurement unit to detect orientation and movement data, NEO-M9N GNSS receiver for acquiring position and speed, and M5StackCAM payload will be used as imaging sensor. All data will be acquired using the flight controller STM32F405, then transmitted to the ground station via the telemetry module SX1276 and simultaneously recorded on a microSD-card.



Payload Subsystem

Electrical Interface

Standard electrical communication buses will be used to connect the payload sensors with the STM32F405 flight controller. Camera will be connected independently, it will work as imaging payload and records the images locally. Power will be provided from 5 V and 3.3 V rails, providing reliable power supply for payload elements.

Mechanical Interface

Mounting provisions were developed for all payload elements to install inside the structure of CAN-7USAT. Such mounting is capable of protecting sensors from external influences, including launch and descent induced forces. Cameras were positioned in such a way that would allow having the unobstructed viewing angle of the horizon line during the descent



Design Constraints and Trade-Offs

There were several design constraints that had to be followed while developing payload subsystem. It included limitations on total weight, dimensions and power consumption. To minimize system complexity and reduce number of components, it was decided to use BMP585, which includes both pressure and temperature sensor. Also, BNO086 was used to acquire motion and orientation data because of integrated sensor fusion, which would reduce data processing load on the flight controller.

Camera payload requires additional power and storage, but it still was used since acquired photos are useful during analysis of mission parameters. Careful consideration was required to position payload components, it was important to distribute payload weight evenly in order to increase aerodynamic stability and have sufficient amount of space in CAN-7USAT.



II. Housekeeping Subsystem



Mechanical subsystem

Housing Mechanical subsystem discuss about:

- Structural enclosure which protects internal components from mechanical damages
- Outer shell which maintains structural integrity during launch and recovery
- Structural Layouts such as top cover, main body, and bottom cover
- Internal structure frame which is hexa pillar system for our CanSat
- Material selection based on mass constraint and strength required by each structure
- Overall design constraint which includes Mass, Volume, Structural and Manufacturability
- Recovery of Cansat and mechanism used to engage the parachutes
- Back-up descent control system
- Parachutes and its shape
- Mass budget of all components

All these are discussed elaborately in the 7.4 Mechanical Subsystem



Communication and Data Handling Subsystem

Key Components and Peripheral Equipment

- Flight control is mainly achieved through the use of a STM32F405 flight computer that manages and monitors all sensor readings, data transmission, and logging operations in an autonomous manner.
- The environmental parameters are measured through a BMP585 sensor that gives both temperature and barometric pressure.
- The orientation is measured by a BNO086 IMU sensor that measures the acceleration, angular velocity, and orientation.
- Localization and time synchronization are carried out using a NEO-M9N GNSS module.
- The ground communication is done via an SX1276 LoRa transmitter.



Communication and Data Handling Subsystem

Data Management and Redundancy

- In order to make sure that data management is executed effectively, the idea of dual path data management has been incorporated into the system.
- During the analysis of the data by the flight computer, the same data gets sent out through the LoRa radio channel to the ground station, as well as being stored in the microSD card in the aircraft itself using the FAT32 file system format.
- The main reason for doing this is because the data cannot be lost even if something goes wrong with the radio.



Reliability of Communication

- The main radio communication channel works on frequencies between 865 and 867 MHz, and it is known for its high reliability.
- The safety margin for this channel was found by link-budget calculations and equals 56 dB in line-of-sight conditions at a distance of 1 kilometer.
- In reality, such margin guarantees the absence of interference with the signal because of the existence of another group and also due to possible tumbling of CanSat during descent.
- This effect can be solved by using an omnidirectional antenna.



System Security and Data Integrity

- The data sampling frequency of the onboard sensors is varied dynamically based on the current stage of flight, with an increase in the data sampling rate to 100 Hz during ascent for capturing rapid dynamic events.
- In order to maintain data integrity and prevent the running of malicious and invalid commands, a very secure four-level validation scheme has been used.
- This validation process involves four steps – hardware checks at the physical level, math checks through checksums, packet counts for any missing data, and a unique team identifier.



Mission Time Continuation and Recovery

- In contrast to relying upon an internal oscillator clock that can drift, the mission time is obtained directly from the GNSS satellite constellation.
- The software used in the flight has been programmed to handle system resets mid-mission smoothly.
- This is achieved by constantly saving pertinent information such as the total mission elapsed time, the number of packets generated since the start of the flight, and the state of the flight itself onto a microSD card.
- This ensures that in case of a computer reset, the saved values are used to continue operation uninterrupted so that the mission remains uninterrupted for the ground station.



Electrical Power Subsystem

- The Electrical Power Subsystem (EPS) of CAN-7USAT is designed to provide stable and reliable power to all onboard electronics while meeting competition requirements. The system uses a 2S1P Samsung INR18650-30Q Li-ion battery pack rated at 7.4 V and 3000 mAh, providing 22.2 Wh of energy. A MP1584 buck converter steps the voltage down to 5 V, while a TPS73733 LDO regulator generates the 3.3 V rail for sensors and communication modules.
- The EPS powers the STM32F405 flight computer, BMP585 altimeter, BNO086 IMU, NEO-M9N GPS, SX1276 LoRa radio, M5Stack camera, and microSD data logger. Protection features include a 3 A fuse, reverse-polarity MOSFET protection, and a TVS diode for surge suppression. Battery voltage is monitored using a resistor divider connected to the STM32 ADC.
- The average system power consumption is approximately 2.49 W. For the required 2-hour mission, the system needs about 4.98 Wh, or 6.47 Wh including a 30% safety margin. With 22.2 Wh available, the EPS provides a theoretical runtime of 8.9 hours and a safe runtime of about 7 hours, comfortably exceeding competition requirements.



Sensor Subsystems

Altitude & Temperature: The BMP581/BMP585 serves as the primary altimeter, chosen for its high accuracy (± 0.5 m) and rapid sampling rate (up to 622 Hz). It utilizes an integrated temperature sensor to eliminate the need for extra modules, while a BMP280 acts as a reliable backup altimeter.

Orientation (IMU): The system relies on the BNO086 IMU rather than standalone magnetometers, as magnetometers suffer from severe interference from high-current motor traces and flight vibrations. The BNO086 features an onboard CEVA MotionEngine to perform advanced sensor fusion and reduce microcontroller load.

Navigation (GNSS): The NEO-M9N provides primary location tracking due to its quad-constellation support and capability to handle aerospace dynamics (up to 80 km altitude and 500 m/s velocity). A NEO-M8N is designated as the secondary fallback.

Telemetry & Storage: RF communication is handled by a 433 MHz LoRa SX1278 transceiver, and onboard logging is managed via an SPI microSD module.



7.4 Mechanical Subsystem



Mechanical subsystem

Mechanical subsystem provides:

- Structural framework
- Safe integration of components
- Recovery System

Major structural components:

1. Outer casing
2. Structural Frame
3. Plates



Structural Components

Outer casing :

- Cylindrical structure conforming to CAN-7USAT dimensions (In our case it is 15x40 CM)
- Protects internal components (electronics) from dust and moisture in air and also protects from Mechanical shocks during ascent and descent

Design Requirements:

- Lightweight
- Accessible
- Ventilation

CanSat outer casing should be light in weight so that it meets the mass budget but, should be strong enough and should have access to remove top and bottom for better integration and ventilation should be provided through vent holes so that sensors like pressure, temperatures exposed to outside air



Structural Components

Material for outer casing:

- PETG
- PLA
- ABS
- Fiberglass
- Carbon fiber

By using 3D-printing we could go with PETG as better option because it is strong and light in weight compared to others

Structural frame

It is the skeleton of the cansat .It supports all components to maintain their position and protect from loads experienced during the launch,payload separation and landing.

Functions:

- Components mounting
- Absorbs loads
- Protection and alignment



Structural components

Structure of frame design:

Number of pillars: 6

We have designed our CanSat with hexapillar system over 3 & 4 pillar system

Reasons:

- Higher structural strength
- Better vibration resistance
- Higher torsional rigidity

Material for structural frame:

- 3D-printed PETG circular plates
- Aluminium pillars

Overall mass of frame of will be around 200-250 grams



Structural Components

Plates:

- Provides mounting surfaces for PCBs,sensors,batteries.
- Increase stiffness for CanSat
- They are Manufactured from Lightweight material either aluminum or composites

Other things such as fasteners and connectors are stainless steel screws ,etc..

Safe integration of components in these structures provides nominal performance of components which meets the mission requirements such as recovery, telemetry,etc..

After separation from Launcher, CanSat needs it's own recovery system to land safely without any damages . CanSat will have two descent control systems(Primary and Secondary)

Recovery options:

- Parachutes
- Rotors
- Gliders
- Streamers



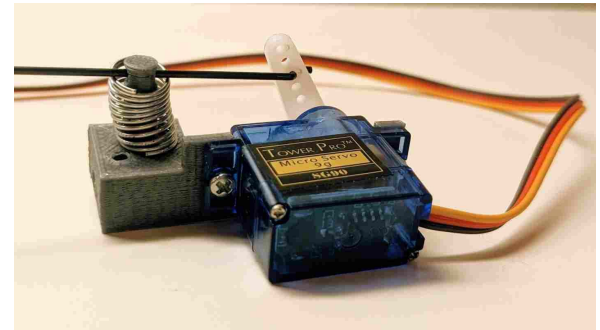
The best options among them are either parachutes or rotors, glider and streamers are not preferable due to mass and volume constraints

Till PDR we have done research on descent control by using parachutes, In CDR phase we want develop rotor mechanism in CanSat to get better descent control and precision landing .

Parachutes deploying mechanism :

Parachutes and spring system are well packed and prepared for launch, Once CanSat is deployed from the rocket, flight controller detect the condition and send command to servo to rotate and latch disengages so that spring push the top part of the CanSat and parachutes will be expelled into airflow and it inflates and continues controlled descent till landing

For primary descent control we use drogue parachutes will Engage at apogee while for secondary descent control system We use main parachutes which will engage at 600m altitude



Backup descent strategy:

All the conditions will be updated to MCU by altitude sensor so, in recovery it plays a vital role in engaging of parachutes at desired altitudes. In case, if any anomaly in sensor part then we activate COTS altimeter which is commercial altimeter, which will monitor the altitude independently and will guide the MCU to take further actions to recover the CanSat by deploying parachutes.

COTS Preference Order:

1. Missile Works RRC3+
2. PerfectFlite StratoLoggerCF
3. EggTimer Quark





About Parachutes

All about parachutes:

Aerodynamics of Parachutes

The drag force is dependant upon

- 1) the **dynamic pressure** created by moving air striking the parachute canopy (and which keeps the parachute inflated)
- 2) the **diameter of the parachute**, which determines the area over which the dynamic pressure acts
- 3) the **drag coefficient, C_d** , of the parachute.

Dynamic pressure is a function of velocity and air density, which in turn is dependent upon altitude and temperature.

Descent rate of parachutes

The vertical descent rate provided by parachute in stable descent rate is given by :

$$V_e = \sqrt{\frac{2 W_t}{S_o C_d \rho}}$$

W_t = total weight of body and parachute

S = canopy reference (surface) area

ρ = air density

To get the canopy area of parachutes :

To get descent velocity around 5m/s the relation between diameter and mass is given by:

$$\text{Diameter} = 0.9 * (\text{mass})^{1/2} \text{ where mass is in kg}$$

Since our CanSat mass is around 1kg , required diameter is around 90cm for main chutes



Calculations for descent velocity

- Mass $m=1\text{kg}$
- Parachute diameter $d=0.9\text{m}$
- Area $A=\pi r^2 =0.64\text{m}^2$
- Drag coefficient $C_d=0.8$
- Air density $\rho=1.225\text{ kg/m}^3$
- Using:

$$v=\sqrt{(2mg/\rho C_d A)} \approx 5.6\text{ m/s}$$

The terminal descent velocity is subjected to change based on air density at location of launch and weather conditions



Parachute design

Types of parachutes based on the shapes:

More common shapes are flat circular, conical, bi-conical, hemispherical, ellipsoidal, cross, annular, toroidal, ringslot, ringsail and ribbon- type

- Parasheet *or* flat circular
- Elliptical *or* ellipsoidal
- Cruciform *or* cross

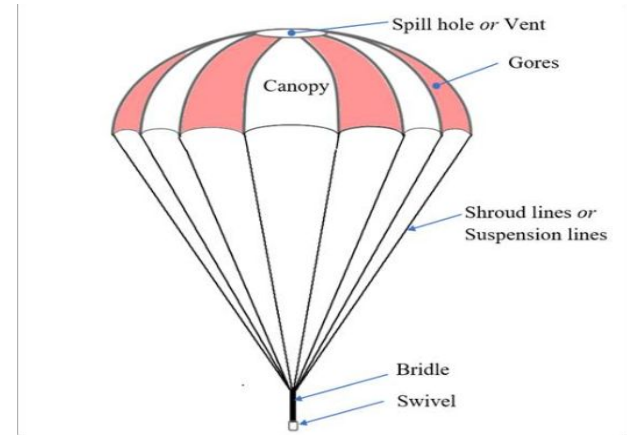
The parasheet is simplest to make, it is basically a single piece of fabric cut to a circular shape. Shroud lines are sewn onto the hemmed circumference.

	Plan view	Profile	Inflated shape	Drag coefficient ^[1]
Parasheet				0.75 to 0.80
Ellipsoidal				0.62 to 0.77
Cross				0.60 to 0.85 ^[2]

According to some research papers,
Circular form parachute is considered best of these

- It will give stable descent rate
- Prevent swinging(oscillations)
- Open reliably

Material: Ripston Nylon





Mass Budget(1/4)

Mass of all electrical components

Component	Model	Estimated Mass (g)
Flight Computer	STM32F405 Flight Controller	15
Altimeter	BMP585 Breakout Module	20
IMU	BNO086 Breakout Module	3
GNSS	NEO-M9N Module + antenna	22
Telemetry Radio	SX1276 LoRa Module	5
Camera	M5StackCAM (OV2640)	15
Data Logger	MicroSD Module + Card	4



Mass Budget(2/4)

Component	Model	Estimated Mass (g)
Buck Converter	MP1584 Module	10
LDO Regulator	TPS73733 Circuit	1
Wiring & Connectors	JSTs, headers, cables	30
PCB (custom avionics board)	Main electronics board	30
Total mass of electrical components		155



Mass Budget(3/4)

Mass of mechanical components

Component	Estimated Mass (g)
2 x Samsung INR18650-30Q	100
Primary Parachute	25
Secondary Parachute	15
Servo Mechanism	30
CanSat Structure (frame, fasteners, mounts)	300
Recovery Buzzer	10
Status LEDs	2
Total mass of mechanical components	482



Mass Budget(4/4)

Total mass budget of CAN7USAT

= Mass of electrical components + Mass of mechanical components
=155 + 482=637g



7.5 Communication and Data Handling(CDH) Subsystem

CDH System Architecture

Architecture Overview

The STM32F405 is the sole flight computer. All sensors, peripherals, and storage are subordinate devices. It manages sensor acquisition, calibration, FSM execution, packet formation, LoRa control, and microSD logging.

Dual-Path Data Flow

Primary Telemetry Path:

Sensors → STM32F405 → Packet Formation
→ SX1276 LoRa → Ground Control Station

Parallel Local Storage Path:

Sensors → STM32F405 → FAT32 microSD

Both paths operate concurrently. Data is preserved even if the LoRa link fails.

M5Stack Camera (OV2640)

Connects via UART. Stores images locally only — does not transmit over LoRa.
Operates at 2.4 GHz, providing RF isolation from the 868 MHz link.

Peripheral Interface Assignments

BMP585 — I²C

PB6 (SCL), PB7 (SDA) | Barometric pressure and temperature

BNO086 — I²C

PB6 (SCL), PB7 (SDA) | IMU: accel, angular velocity, orientation

NEO-M9N — UART1

PA9 (TX), PA10 (RX) | GNSS positioning and UTC time

SX1276 — SPI2

PB13(SCK) PB14(MISO) PB15(MOSI) PB12(NSS) | LoRa transceiver

microSD — SPI1

PA5(SCK) PA6(MISO) PA7(MOSI) PA4(CS) | FAT32 local storage

M5Stack Camera — UART

Pins TBD (confirmed at hardware integration) | Image capture

BMP585 and BNO086 share the I²C bus; each uses its unique device address.

Communication Protocol — SX1276 LoRa

LoRa Configuration Parameters

Operating Band: 865–867 MHz (868 MHz nominal)

Bandwidth: 125 kHz

Spreading Factor: SF7

Coding Rate: 4/5

TX Power: 17 dBm

Preamble: 8 Symbols | **Sync Word:** 0x12 (Private Network)

CRC: Enabled | **Mode:** Half-duplex | **Rate:** 1 Hz



Design Rationale

SF7 at 125 kHz prioritises throughput and reduced air time. Link budget analysis confirms 56 dB margin at 1 km LOS — substantially exceeding operational requirements.

The private sync word 0x12 isolates the CanSat link from other LoRa devices at shared launch events.

Hardware CRC provides physical-layer error detection. This is the first of four independent integrity layers in the system.

Half-duplex uses dedicated TX/RX windows. The CanSat enters LAUNCH_PAD on power-up and awaits CMD_START_TELEMETRY before activating the downlink, preventing premature RF emissions.

Telemetry Packet Content (≈73 bytes, fixed layout)

- Mission Time, Packet Count, Flight State
- Altitude, Pressure, Temperature
- Battery Voltage, System Health Flags
- GPS: latitude, longitude, altitude, satellite count
- IMU: acceleration X/Y/Z, angular velocity X/Y/Z, orientation

Fixed-length binary packed C structure — deterministic GCS parsing.



Communication Link Budget Analysis

Step 1 — Free-Space Path Loss

Signal attenuation due to geometric spreading over the link distance (unobstructed LOS):

$$\text{FSPL(dB)} = 20 \cdot \log_{10}(d) + 20 \cdot \log_{10}(f) + 32.44$$

d = distance in km, f = frequency in MHz

$$\begin{aligned}\text{FSPL} &= 20 \cdot \log(1) + 20 \cdot \log(867) + 32.44 \\ &= 0 + 58.76 + 32.44 \approx 91 \text{ dB}\end{aligned}$$

Step 2 — Received Power

$$\begin{aligned}P_r &= P_t + G_t + G_r - \text{FSPL} \\ P_t &= 17 \text{ dBm} \quad (\text{TX power}) \\ G_t &= 2 \text{ dBi} \quad (\text{CanSat monopole}) \\ G_r &= 5 \text{ dBi} \quad (\text{GCS collinear})\end{aligned}$$

$$P_r = 17 + 2 + 5 - 91 = -67 \text{ dBm}$$

Step 3 — Link Margin

$$\begin{aligned}\text{LM} &= P_r - \text{Sensitivity} \\ &= -67 - (-123) = 56 \text{ dB}\end{aligned}$$

What Does 56 dB Mean?

The received signal is 56 dB above the SX1276 demodulation floor. Every 6 dB doubles tolerable degradation:

- Antenna orientation loss (monopole perpendicular to link): 10–15 dB — fully covered
- Typical multipath fading: 10–20 dB — fully covered
- Launch-site interference from adjacent LoRa teams: covered
- Conservative values used (2 dBi / 5 dBi) — actual margin likely higher

SF7 confirmed appropriate for 1 km LOS profile. Adjustable to higher SF if ground testing reveals unexpected degradation.

TX: 17 dBm | CanSat: 2 dBi | GCS: 5 dBi

FSPL @ 1 km, 867 MHz $\approx$ 91 dB | P_r = -67 dBm

Sensitivity: -123 dBm | Link Margin: 56 dB — Robust

Antenna System Design

CanSat — Quarter-Wave Monopole

Omnidirectional azimuthal radiation pattern selected because CanSat orientation cannot be controlled during descent.

Length Calculation:

$$L = c / (4 \times f)$$

$$c = 3 \times 10^8 \text{ m/s}, \quad f = 867 \text{ MHz}$$

$$L = (3 \times 10^8) / (4 \times 867 \times 10^6) \approx 8.6 \text{ cm}$$

Copper conductor selected for high conductivity, low mass, and ease of fabrication. PCB ground plane serves as counterpoise.

CanSat Antenna Specifications

- Type: Quarter-wave whip monopole
- Material: Copper wire conductor
- Frequency: 865–867 MHz
- Gain: 2–3 dBi
- Polarisation: Vertical
- Pattern: Omnidirectional (azimuthal)
- Length: $\approx 8.6 \text{ cm}$
- Mounting: Above PCB ground plane (counterpoise)



GCS Antenna

Fiberglass-enclosed collinear, selected for increased gain while maintaining omnidirectional coverage without active tracking. Compensates for the CanSat's constrained installation and contributes directly to the 56 dB link margin.

GCS Antenna Specifications

- Type: Fiberglass omnidirectional collinear
- Material: Radiating elements enclosed in fiberglass radome
- Frequency: 865–867 MHz
- Gain: 5–8 dBi
- Polarisation: Vertical
- Pattern: Omnidirectional
- Mounting: Fixed tripod or mast
- Tracking: Not required



Polarisation Matching

Both antennas use vertical polarisation. Matched polarisation minimises polarisation loss — a mismatch would contribute 3–6 dB of avoidable attenuation.

Why No Tracking?

The 56 dB margin provides sufficient tolerance for signal variation caused by CanSat rotation over the 1 km mission profile. Active tracking hardware adds mass and complexity without a measurable benefit given the available margin.



Telecommand Architecture — Uplink Design

6-Byte Command Packet

Byte 0: Sync Byte — 0xAA

Bytes 1–2: Team Number (uint16_t)

Byte 3: Opcode — command identifier

Byte 4: Parameter — command argument

Byte 5: XOR Checksum over Bytes 0–4

ACK / NACK Protocol

Valid command (all layers passed + FSM state compatible) → ACK.

Invalid/incompatible → NACK. Distinguishes a rejected command from a lost packet — diagnostic information a timeout-only approach cannot supply.

Supported Commands

- 0x01 CMD_START_TELEMETRY
- 0x02 CMD_STOP_TELEMETRY
- 0x03 CMD_CALIBRATE_BARO
- 0x04 CMD_CALIBRATE_GYRO
- 0x05 CMD_CALIBRATE_ACCEL
- 0x06 CMD_ENTER_TEST
- 0x07 CMD_ENTER_LAUNCHPAD
- 0x08 CMD_SWITCH_CHANNEL

3-Layer Validation

Layer 1 — Sync Byte (0xAA)

Rapid pre-filter. Discards malformed packets, truncated transmissions, and frames from unrelated LoRa devices. First position minimises rejection cost.

Layer 2 — Team Number

At a shared launch event, multiple teams operate LoRa GS on the same frequency. The uint16_t team identifier is a logical address layer preventing cross-team command interference.

Layer 3 — XOR Checksum

XOR over Bytes 0–4. Receiver independently recomputes and compares. Any bit corruption causes a mismatch and packet rejection. Second independent integrity check above the LoRa hardware CRC.

The three layers defend against: noise-induced bit errors, cross-team interference, and corrupted packet execution. No single layer is sufficient in isolation.



Data Acquisition System

BMP585 — Barometric Pressure Sensor

Provides atmospheric pressure and temperature. Altitude derived via barometric formula. Primary sensor for flight state detection and apogee identification.

- I²C on PB6/PB7; unique device address
- Power: 0.000016 W continuous

BNO086 — Inertial Measurement Unit

3-axis acceleration, angular velocity, and fused orientation (roll, pitch, yaw). Onboard fusion processor reduces STM32F405 load.

- I²C on PB6/PB7; shares bus with BMP585
- Power: 0.059 W continuous

NEO-M9N — GNSS Receiver

GPS positioning (lat, lon, alt) and UTC time. UTC drives mission elapsed time — absolute reference independent of onboard crystal.

- UART1 on PA9 (TX) / PA10 (RX)
- Power: 0.224 W continuous

Adaptive Sampling Strategy

BMP585 — Launch pad: 5 Hz

Low dynamics; baseline monitoring

BMP585 — Ascent: 20 Hz

Sufficient resolution for apogee detection

BNO086 — Initialisation: 10 Hz

Calibration and baseline establishment

BNO086 — Ascent: 100 Hz

Captures separation, deployment, and attitude transients

NEO-M9N — All phases: 1–5 Hz

Bounded by receiver dynamics; adequate for position tracking

Telemetry Downlink — All phases: 1 Hz

Data aggregated from higher-rate sensors; LoRa downlink rate

100 Hz IMU rate during ascent is the key design decision — capturing high-frequency dynamic events that would be aliased at lower rates.



Flight State Machine — 7-State Mission FSM

FSM State Transitions

BOOT → TEST_MODE

Completion of initialisation sequence

TEST_MODE → LAUNCH_PAD

Successful subsystem validation (all health checks passed)

LAUNCH_PAD → ASCENT

Sustained acceleration above configurable threshold AND positive altitude rate

ASCENT → PRIMARY_DESCENT

Apogee detected AND sustained negative vertical velocity

PRIMARY_DESCENT → SECONDARY_DESCENT

Altitude below 600 ± 10 m AGL [only finalised threshold at PDR stage]

SECONDARY_DESCENT → IMPACT

Near-ground altitude AND IMU-detected deceleration spike

IMPACT — Terminal State

Recovery mode and data preservation active

Configurable Threshold Philosophy

Only 600 ± 10 m AGL is finalised (mission specification). All others are intentionally deferred:

- Thresholds depend on aerodynamic and structural characteristics of the final assembly not reliably predictable without empirical data
- Premature commitment risks assumptions invalidated by hardware changes or parachute behaviour in testing
- Configurable parameters → threshold updates are a configuration activity, not firmware recompilation

Qualification Process

- 6-DOF simulation to establish initial estimates
- Bench testing of individual sensor step responses
- Integrated ground testing of full FSM
- Flight validation and refinement
- All finalised values in CDR

Boot Sequence

Power → Sensor init → Bus init (SPI1, SPI2, I²C, UART1) → SD mount → Telemetry init → Health check → State restore → BOOT → TEST_MODE → LAUNCH_PAD (awaits CMD_START_TELEMETRY)



CDH Subsystem Power Budget

Calculation Method: $P = V \times I$

$$P = V \times I$$

P = power (W), V = voltage (V), I = current (A)

Example — STM32F405:

$$V = 3.3 \text{ V}, \quad I = 110 \text{ mA} = 0.110 \text{ A}$$

$$P = 3.3 \times 0.110 = 0.363 \text{ W}$$

All values derived by the same method using datasheet figures.

SX1276 Duty Cycle Analysis

Half-duplex at 1 Hz: TX occupies $\approx 20\%$ of each cycle; idle 80%.

$$\text{TX avg} = 0.396 \text{ W} \times 0.20 = 0.079 \text{ W}$$

$$\text{Idle avg} = 0.033 \text{ W} \times 0.80 = 0.026 \text{ W}$$

$$\text{Weighted} \approx 0.105 \text{ W}$$

M5Stack Camera

At 0.80 W it is the largest single CDH consumer. Intermittent duty cycle (descent phase only) limits its total energy contribution to the 2-hour mission budget.

CDH Power Consumption Table

Component	Voltage	Current	Power	Duty
STM32F405	3.3 V	110 mA	0.363 W	Continuous
BMP585	3.3 V	0.005 mA	0.000016 W	Continuous
BNO086	3.3 V	17.9 mA	0.059 W	Continuous
NEO-M9N	3.3 V	67.9 mA	0.224 W	Continuous
SX1276 TX	3.3 V	120 mA	0.396 W	20% duty
SX1276 Idle	3.3 V	10 mA	0.033 W	80% duty
microSD	3.3 V	50 mA	0.165 W	Continuous
M5Stack Camera	5.0 V	160 mA	0.800 W	Intermittent

Energy budget calculated for 2-hour mission duration.



Data Processing, Integrity and Storage

Onboard Processing

STM32F405 handles all processing: sensor reading, unit conversion, calibration, FSM execution, and packet assembly. Active FSM state governs sampling rates and behaviour.

Packet Formation

Fixed-layout binary packed C structures, ≈73 bytes. Deterministic GCS parsing, no framing ambiguity.

Data Integrity — 4 Layers

Layer 1 — LoRa Hardware CRC

Physical-layer bit-error detection by SX1276 hardware on every frame.

Layer 2 — Application XOR Checksum

Computed over telemetry payload before TX. Verified on reception. Failed packets rejected before processing.

Layer 3 — Sequential Packet Counter

Incrementing counter per packet. Gap detection and continuity monitoring at GCS.

Layer 4 — Telecommand Validation

Sync byte + Team ID + XOR on uplink. ACK/NACK closes the command loop.

Storage Architecture

FAT32 microSD on SPI1 (PA5–PA7, PA4 CS). Captures:

- Telemetry data — all sensor readings and values
- Mission state transitions — FSM state change log
- Recovery information — GPS position at landing

Mission Continuity After Reset

Persisted continuously to microSD:

- Mission elapsed time
- Packet counter
- Current flight state

On reset, values restored before resuming. Preserves packet numbering continuity, mission time coherence, and state consistency.

Mission Timing

NEO-M9N GPS UTC time provides absolute time reference independent of the onboard crystal oscillator, eliminating drift errors.



7.6 Electrical Power Subsystem (EPS) Design



EPS Overview(1/4)

CanSat component summary

Category	Component	Function/Role	Operating Voltage
Flight computer	STM32F405	Main MCU-telemetry,sensor fusion,mission control	5V
Altimeter	BMP585	High precision barometric altimeter+temperature	3.3V
IMU	BNO086	9-axis orientation,acceleration,altitude estimation	3.3V
GNSS	NEO-M9N	Quad-constellation positioning & velocity	5V
Telemetry radio	SX1276 LoRa	Long-range RF communication-868MHz	3.3V
Camera	M5StackCAM(OV2640)	Mission imaging & video capture	5V
Data logger	MicroSD module	SPI flight data logging	3.3V
Buck Converter	MP1584	7.4 V → 5 V, 3 A max, 90–92 % efficiency	Input 7.4V
LDO Regulator	TPS73733	5 V → 3.3 V, 1 A, low-noise high-PSRR	Input 5V



EPS Overview(2/4)

Category	Component	Function/Role	Operating voltage
Surge Protection	SMBJ12A TVS Diode	Transient Voltage clamp-12V, 600 W peak	—
Voltage measurement	R-Divider 100k/47k+ADC	Battery voltage → STM32 PA0 ADC	—
Power Control	External Toggle Switch	Accessible main power ON/OFF	—
Power Indicator	Green LED+330Ω	Power ON visual indicator → PC13	3.3V



EPS Overview(3/4)

EPS Architecture

Battery:	Samsung INR18650-30Q — 2S1P — 7.4V / 3000 Ahm
Main 5V Rail:	MP1584 Buck: 7.4V → 5V, 3 A, 90 % efficiency
Logic 3.3V Rail:	TPS73733 LDO: 5V → 3.3V, 1A, low-noise
Architecture:	Dual regulated voltage, centralised star distribution
Protection:	Fuse + MOSFET reverse polarity + TVS surge clamp
Mission Runtime:	2 h minimum required ~7h safe operational
Total Energy:	22.2 Wh available 5.64 Wh required (+30 % margin)

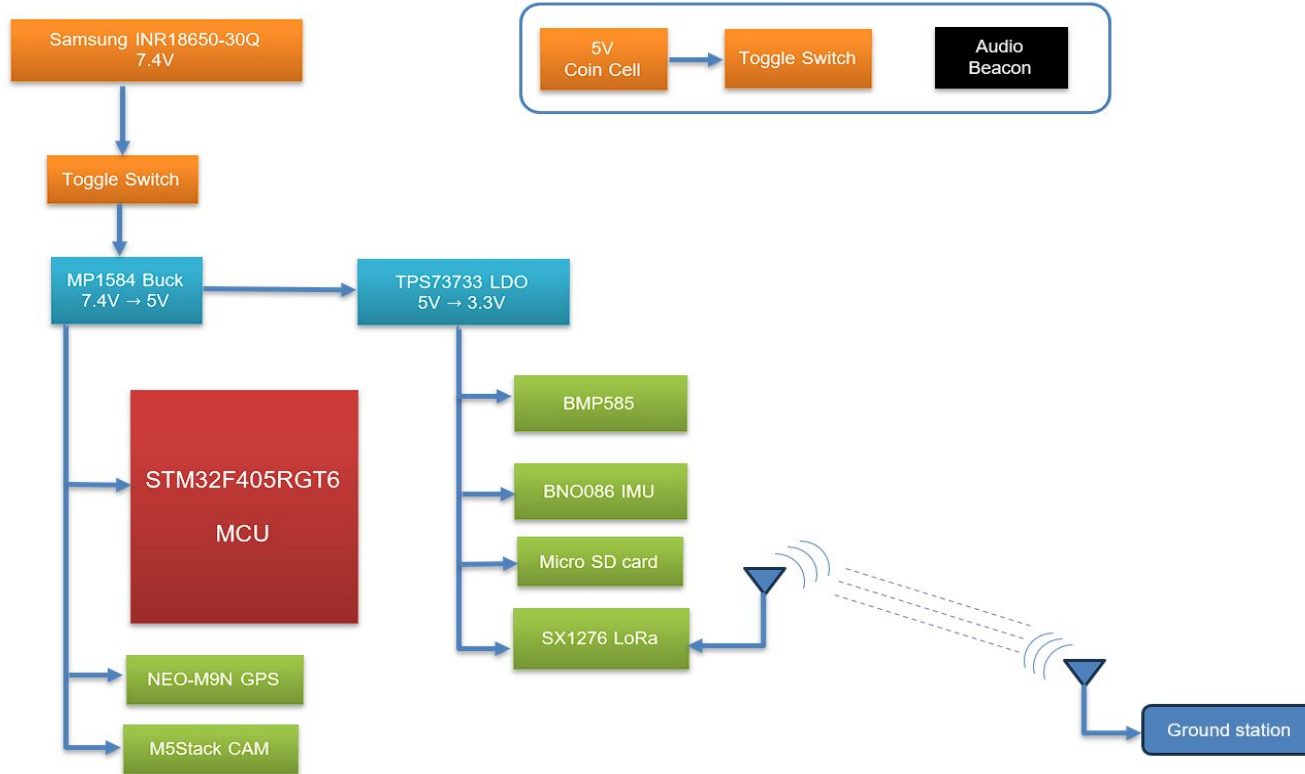


EPS Overview(4/4)

Requirements Compliance

Requirement	Status
External accessible power switch	YES
2-hour minimum operation	YES(~7h actual)
Cylindrical metal-cell battery	YES–INR18650
Fuse protection	YES–3 A blade fuse
Surge/transient protection	YES–SMBJ12A TVS
Voltage monitoring	YES–voltage divider → ADC PA0

Payload Electrical Block Diagram



Battery Type Trade & Selection(1/3)

Parameter	Samsung INR18650-30Q	Panasonic NCR18650GA	Molicel INR21700-P45B
Cell Type	Li-Ion INR18650	Li-Ion NCR18650	Li-Ion INR21700
Nominal Voltage	3.6 V	3.6 V	3.6 V
Full Charge	4.2 V	4.2 V	4.2 V
Capacity	3000 mAh	3500 mAh	4500 mAh
Energy / Cell	10.8 Wh	12.6 Wh	16.2 Wh
Max Cont. Current	15 A	10 A	45 A
Weight / Cell	48 g	48 g	70 g
Size	18.4 x 65 mm	18.6 x 65 mm	21 x 70 mm
Thermal Stability	Very Good	Excellent	Excellent
Cost / Cell	INR 550–650	INR 700–850	INR 1000–1200



Battery Type Trade & Selection(2/3)

Model Selected: Samsung INR18650-30Q

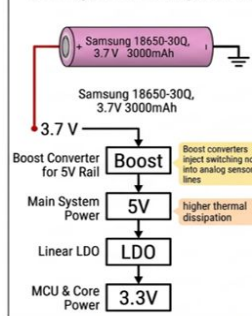
- **Competition Legal — cylindrical metal cell (INR18650)**
- **High current capability — 15 A continuous per cell**
- **Lightweight — 48 g/cell vs 70 g (Molicel P45B)**
- **High availability in India — widely stocked locally**
- **Rechargeable — Li-Ion chemistry, reusable**
- **Supports camera (M5Stack) and LoRa TX burst loads**

Battery Configuration Trade & Selection(3/3)

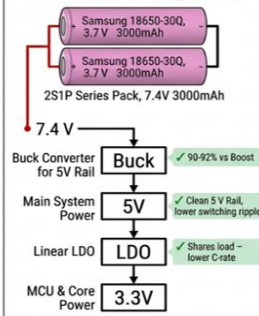
Battery configuration selection

Configuration	Nominal Voltage	Capacity	Power flow	Max Discharge Power	Verdict
1S Single Cell	3.7 V	3000 mAh	3.7 V → Boost → 5V → LDO → 3.3V	9.25 W	✗ REJECTED
2S1P Series	7.4 V	3000 mAh	7.4 V → Buck → 5V → LDO → 3.3V	22.2 W	✓ SELECTED
2P Parallel	3.7 V	6000 mAh	3.7 V → Boost → 5V → LDO → 3.3V	18.5 W	✗ REJECTED

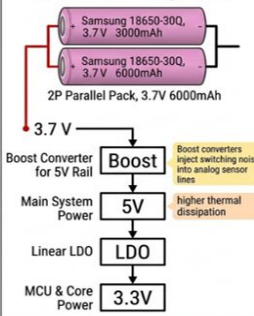
1S Single Cell Configuration



2S1P Series Configuration



2P Parallel Configuration



Selection:

- **Buck converter efficiency** 7.4V → 5V buck achieves 90-92% vs Boost from 3.7V at 80-85%
- **Clean 5V rail** Direct step-down — lower switching ripple for camera and LoRa
- **Lower I2C/ADC Noise** Boost converters inject switching noise into analog sensor lines
- **Lower cell current stress** 2S shares load — lower C-rate, less heat, longer cell life
- **Thermal Safety** Buck has lower thermal dissipation than boost at same load



Power Budget (1/2)

Type	Component	Voltage(V)	Current(mA)	Power(W)	Duty Cycle(%)	Energy 2h(Wh)	Source
MCU	STM32F405	3.3	110	0.363	100	0.726	Datasheet
Sensor	BMP585	3.3	0.005	0.000016	100	0.00003	Datasheet
IMU	BNO086	3.3	18	0.059	100	0.118	Datasheet
GPS	NEO-M9N	3.3	68	0.224	100	0.448	Datasheet
Radio	SX1276 LoRa TX	3.3	120	0.396	20	0.158	Datasheet
Radio	SX1276 LoRa Idle	3.3	10	0.033	80	0.053	Datasheet
Storage	microSD Module	3.3	50	0.165	100	0.330	Datasheet
Camera	M5Stack CAM	5.0	160	0.800	100	1.600	Datasheet
LED	Power LED	3.3	10	0.033	100	0.066	Estimated
Loss	Reg. & Conv. Losses	—	—	0.420	100	0.840	Calculated
TOTAL				2.49		4.339	



Power Budget (2/2)

Battery Energy Analysis

Battery Energy = Voltage $\times$ Capacity = 7.4 V $\times$ 3.0 Ah	= 22.2 Wh
Total Power Consumption (avg)	= 2.49 W
Mission Energy Required (2 h) = 2.49 W $\times$ 2 h	= 4.98 Wh
Safety Margin (30 %)	= 6.47 Wh (final required)
Energy Margin = 22.2 - 6.47	= 15.73 Wh
Operating Time = 22.2 / 2.49	= 8.91 Hours (theoretical)
Safe Runtime	» 7 Hours (derated: temp + aging + RF bursts)

Battery Pack Energy	Mission Energy Budget	Margin Analysis
Total Available Energy = 22.2 Wh	Avg. Power Consumption = 2.49 W	Margin Percentage = 243 %
Battery Voltage = 7.4 V (2S)	2h Mission Energy = 4.34 Wh	Safe Runtime ~7 Hours
Capacity = 3000 mAh	With 30% Safety Margin = 6.47 Wh	Competition Minimum = 2 Hours
Theoretical Runtime of battery = 8.91 h	Energy Remaining = 15.73 Wh	Status = COMPLIANT ✓



Component Cost Budget(1/2)

Approximate market prices verified from Indian suppliers: **Robu.in, Robocraze and ElectronicsComp.**

Component	Qty	Unit Cost (INR)	Total (INR)
STM32F405 Development Board	1	850	850
BMP585 Sensor Module	1	690	690
BNO086 IMU Module	1	2100	2,100
NEO-M9N GPS Module	1	2350	2,350
SX1276 LoRa Module	1	420	420
M5Stack Unit CAM (OV2640)	1	1150	1,150
microSD SPI Module	1	120	120
MP1584 Buck Converter Module	1	95	95
TPS73733 LDO Regulator	1	180	180



Component Cost Budget(2/2)

SMBJ12A TVS Diode	1	30	30
XT60 Connector Pair	1	80	80
3A Automotive Fuse	1	25	25
Samsung INR18650-30Q Cells	2	625	1,250
Battery Holder + Nickel Strip	1	220	220
External Toggle Switch	1	60	60
Power LED + Resistor	1	10	10
Capacitors + Ferrites + Passives+Wiring + Connectors + Mounts	—	—	180+450
SUBTOTAL+Contingency(10%)			INR 10,300+INR 1,210
TOTAL ESTIMATED EPS BUDGET			INR 11,510



7.7 Sensor Subsystem Details



Sensor Subsystem Overview

Subsystem	Primary Sensor	Function
Altitude+Temperature	BMP581	Provide integrated temperature readings, eliminating the need for a separate temperature module.
Air Temperature	BMP581 (Integrated)	
Navigation (GNSS)	NEO-M9N	Measure absolute position, velocity, and time data throughout the rocket's flight
Tilt Orientation / IMU	BNO086	To track the rocket's orientation, rotation rate, and linear acceleration
Magnetometer	BNO086 (via fusion)	Measure the direction and magnitude of the Earth's magnetic field, CEVA fusion, real-time orientation
Telemetry	LoRa SX1278	RF 433 MHz
Data Logging	microSD (SPI)	—

Payload Air Pressure Sensor Trade & Selection

Sensor	Weight	Pressure Accuracy	Altitude Accuracy	Communication Interface	Output Data Rate (ODR)	Power Consumption	Operating Voltage	Cost (INR)
BMP180	5g	$\sim \pm 1-4$ hPa	~ 0.03 hPa (18-bit)	I2C only	Low (~ 1 Hz)	~ 650 μ A	1.8–3.6V	$\sim ₹50$
BMP280	3-5g	$\sim \pm 1$ hPa @ 25°C	± 0.12 hPa (± 1 m)	I2C, SPI	Moderate	~ 714 μ A	1.71–3.6V (3.3V module)	$\sim ₹400-800$
BMP581 / BMP585	3-5g	± 0.5 hPa max	± 6 Pa ($\sim \pm 0.5$ m)	I2C, I3C, SPI	Up to 622 Hz	~ 260 μ A active	1.71–3.6V (3.3V module)	$\sim ₹800-1100$

Critical Performance Requirements

- Altitude resolution: < 1 m (required for apogee detection)
- Sampling rate: 100–622 Hz (fast ascent tracking)
- Noise immunity: Critical (high vibration environment)
- Shock survival: Passive MEMS OK with foam/rubber damping
- Temperature: Onboard compensation preferred for flight accuracy





Payload Air Temperature Sensor Trade & Selection

Role	Sensor	Justification
Primary Altimeter	BMP581 / BMP585	622 Hz ODR, ± 0.5 m accuracy, FIFO, onboard LPF — best-in-class for rocketry
Backup Altimeter	BMP280	Proven, low-cost redundancy; adequate accuracy for failover
Temperature	BMP581 (Integrated)	Use internal sensor — tightly calibrated, no extra weight/cost

Payload GNSS Sensor Trade & Selection

Sensor	Weight	Position Accuracy	Satellite Constellations	Communication Interface	Output Data Rate (ODR)	Power Consumption	Operating Voltage	Cost (INR)
NEO6M	50g	2.5m	GPS only	UART	Low (~1 Hz)	Low	2.7–3.6V	~ ₹300-500
NEOM8N	25g	0.6-0.9m	GPS+GLONAS only	UART	Moderate	Low	3.3V	₹2000
NEO M9N	Light	1.5m CEP	GPS+GLONASS+Galileo+Beidou	UART, I2C, SPI	Up to 622 Hz	Moderate(30μA)	3.3V	~ ₹2000+

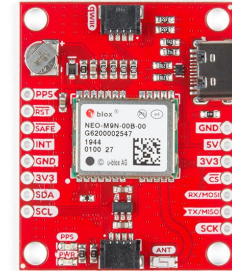
Sensor Justification:

Primary- GPS NEO-M9N:

25 Hz, quad-constellation, anti jamming, 80 km altitude, 500 m/s
— rated for aerospace dynamics

Secondary Backup- GPS NEO-M8N

High accuracy (0.9 m), proven in UAVs/drones, lower cost fallback



Payload Auto-Gyro Rotation Sensor Trade & Selection

Sensor	Accel Range	Sensor Fusion	Output Data Format	Communication Interface	Gyro Range	Power Consumption	Operating Voltage	Cost (INR)
MPU-6050	2-16g	DMP(built onbo	Raw+DMP format	I2C	250-2000 dps	Low	2.3–3.4V	~ ₹150-250
BNO055	2-16g	Built in Fusion	Euler, Quaternion	I2C	250-2000 dps	Low	2.4-16V	₹1300-170]
BNO086	2-16g	Adv motionengine	Orientation, rotation vectors	I2C, SPI	High precision	Moderate(30μA)	3.3V	~ ₹2000+

BNO086-

- CEVA Motion Engine: advanced onboard sensor fusion
- Real-time orientation output with dynamic calibration
- Handles vibration and high-motion environments better than BNO055
- Minimizes MCU load — direct quaternion/rotation vector output
- Designed for robotics, drones, and high-motion aerospace systems



Payload Tilt Sensor Trade & Selection

Interference Source	Impact on Magnetometer
Motor/Igniter Current (high A)	Generates strong local magnetic field — overwhelms Earth's field
PCB Power Traces	High-current traces create localized interference
Vibration (15–30G)	Mechanical stress induces sensor noise
Rapid Rotation	Fast attitude changes reduce sensor response accuracy
Result	Magnetometer readings UNRELIABLE during powered flight

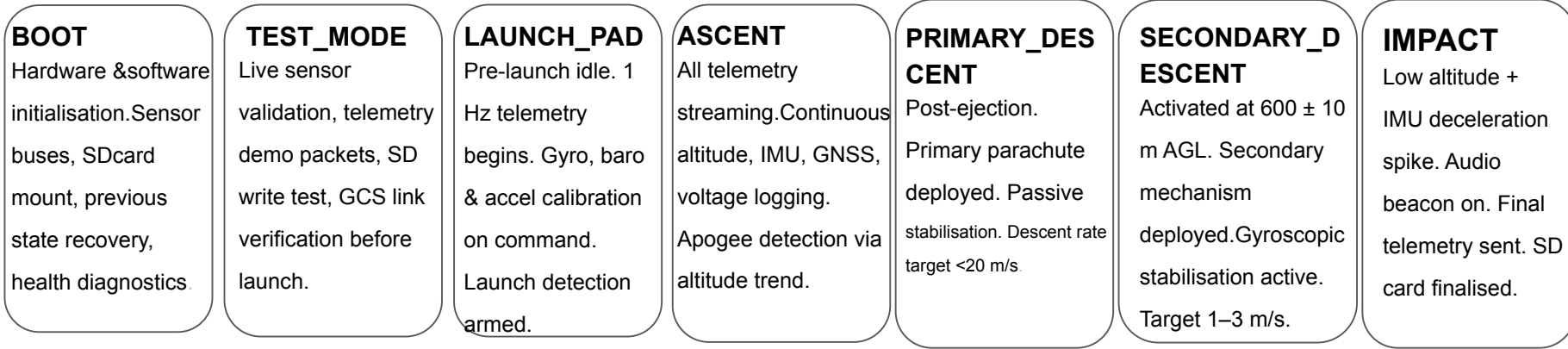
NOTE: After analysing various magnetometers, a conclusion was made to not rely on standalone magnetometers for flight control in rocketry. Instead, use Gyroscope as primary attitude sensor, Accelerometer for support Barometer for altitude, and GPS for position.

FINAL SENSOR COMBINATION	Sensor	Justification
Primary Orientation	BNO086(IMU)	Fusion IMU eliminates magnetometer noise issues; best practical solution
Backup Mag	MMC5983MA	Use only if implementing advanced sensor fusion; $\pm 0.5^\circ$ accuracy, 1000 Hz ODR
Temperature	BMP581 (Integrated)	Use internal sensor — tightly calibrated, no extra weight/cost



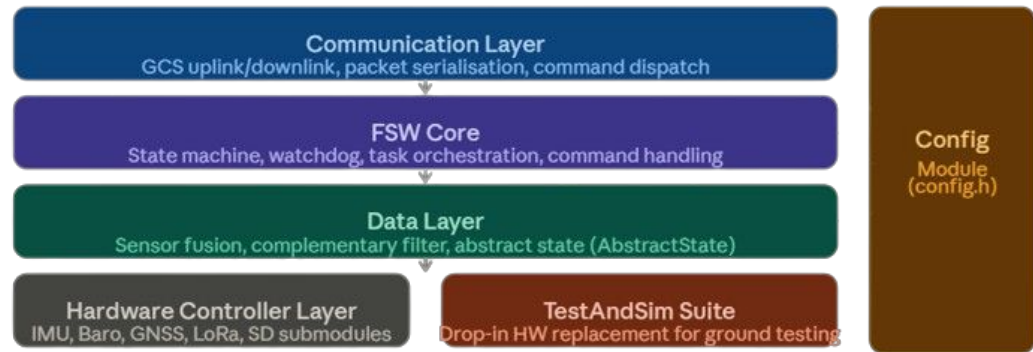
Flight Software (FSW) Design

FSW Overview(1/2)



The finite state machine contains the above states.

The FSW was divided into the following modules by functionality





FSW Overview(2/2)

FSW TASKS (Scheduled by FreeRTOS)

- TaskIMU
- TaskBaro
- TaskGNSS
- TaskDataLayer
- TaskFSM
- TaskTelemetry
- TaskCommRx
- TaskSDWrite
- TaskHealthMonitor
- TaskRecoveryBeacon

Platform

- MCU: Arduino Nano 33 BLE Sense Rev2
ARM Cortex-M4F @ 64 MHz · C++ /
PlatformIO + VSCode
- Radio: XBee PRO S3B 900 MHz
- NETID/PANID = 3165 · 1 Hz telemetry
downlink
- Storage: SD card + processor flash memory
- State and telemetry persisted across resets
- RTC: DS3231 mini
Maintains mission time through power loss

Task Sampling Rate Across States

CAN-7USAT FSW - Decided Sampling Rates Across FSM States

Task	Module	BOOT	TEST	LAUNCH_PAD	ASCENT	PRIMARY_DESCENT	SECONDARY_DESCENT	IMPACT
TaskIMU	HardwareController	10 Hz	10 Hz	10 Hz	100 Hz	50 Hz	100 Hz	10 Hz
TaskBaro	HardwareController	5 Hz	5 Hz	5 Hz	20 Hz	10 Hz	20 Hz	2 Hz
TaskGNSS	HardwareController	1 Hz	1 Hz	5 Hz	5 Hz	5 Hz	5 Hz	1 Hz
TaskDataLayer	DataLayer	5 Hz	5 Hz	5 Hz	20 Hz	10 Hz	20 Hz	2 Hz
TaskFSM	FSWCore	10 Hz	10 Hz	10 Hz	10 Hz	10 Hz	10 Hz	10 Hz
TaskHealthMonitor	FSWCore	1 Hz	1 Hz	1 Hz	1 Hz	1 Hz	1 Hz	1 Hz
TaskTelemetry	CommLayer	off	off	1 Hz*	1 Hz	1 Hz	1 Hz	final TX
TaskCommRx	CommLayer	off	off	intr.	intr.	intr.	intr.	off
TaskSDWrite	HardwareController	1 Hz	1 Hz	1 Hz	1 Hz	1 Hz	1 Hz	finalise
TaskComm4G	HardwareController	dormant	dormant	dormant	dormant	dormant	dormant	once

* TaskTelemetry activates on CMD.START.TELEMTRY from GCS, expected during LAUNCH.PAD.

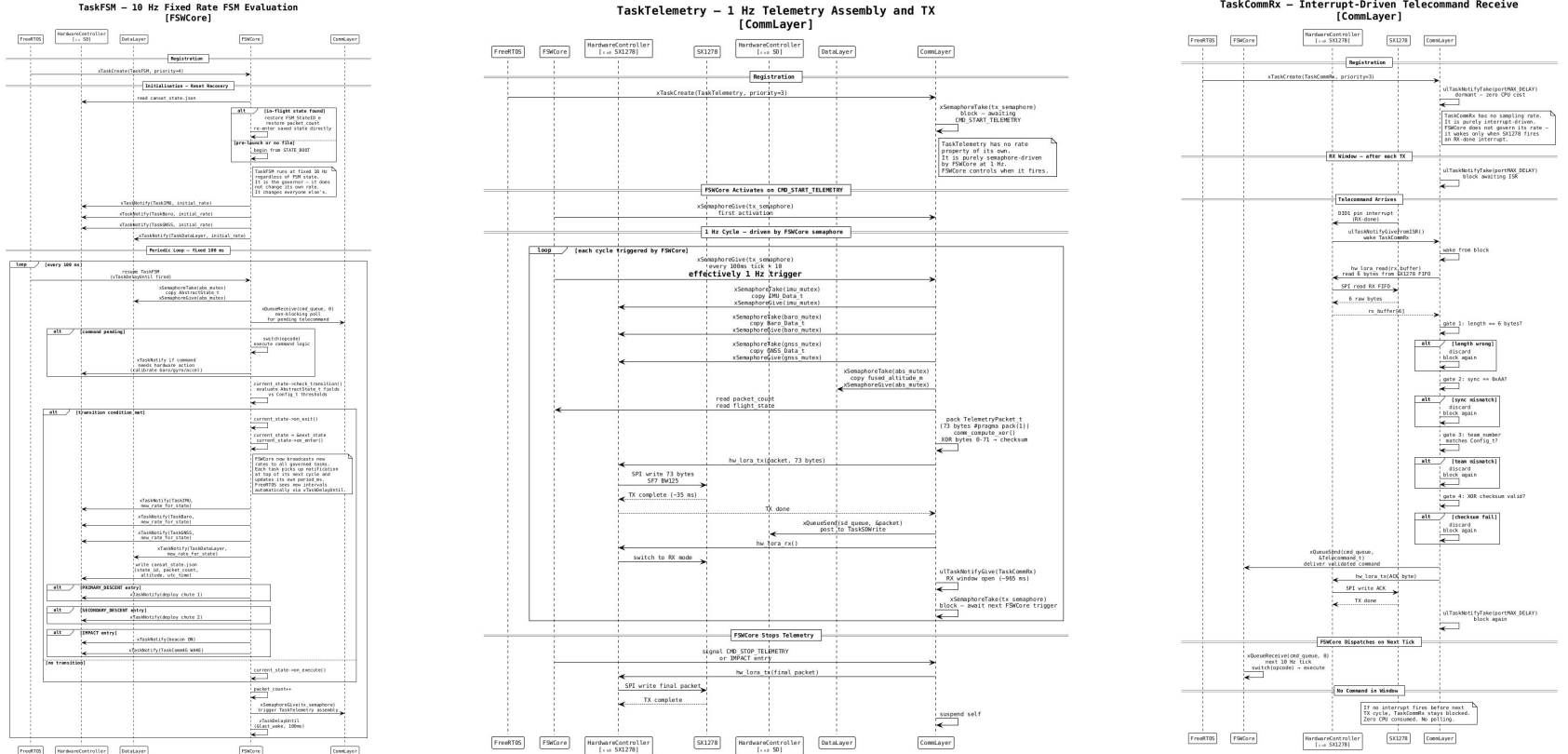
* TaskFSM and TaskHealthMonitor hold constant rate across all states.

* all other periodic tasks updated by FSWCore via xTaskNotify on each state transition.

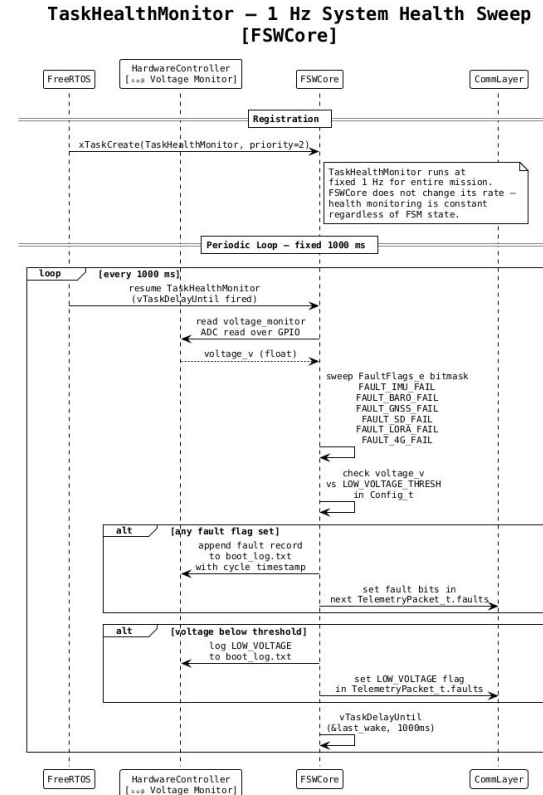
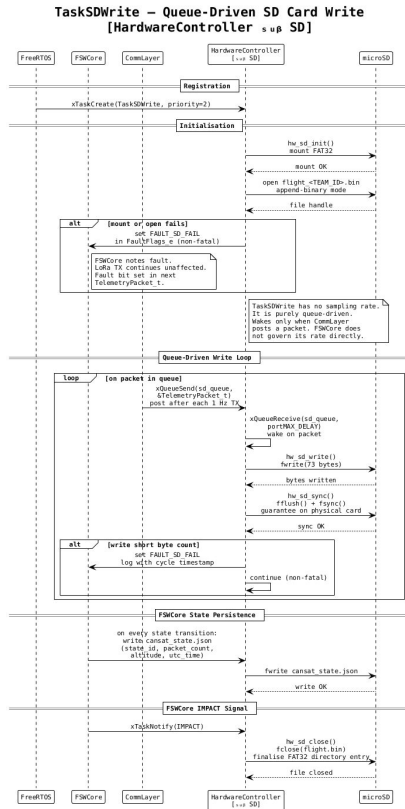
* interrupt driven (SX1278 RX-done ISR), no fixed rate.



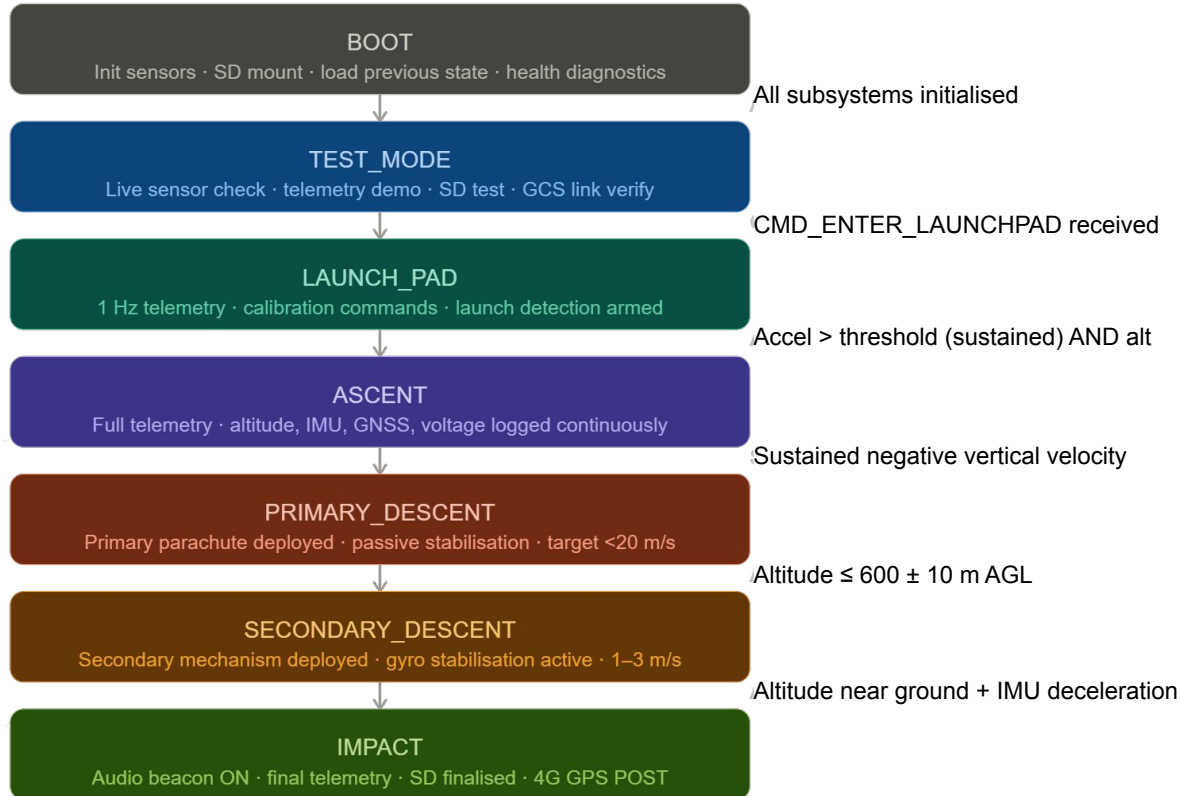
Software Sequencing Diagrams (2/3)



Software Sequencing Diagrams (3/3)



Payload FSW State Diagram (1/2)



Onboard sensor suite

BMP586 — pressure / altitude / temperature (I²C/SPI)
 NEO-M9N — GNSS position, altitude, UTC (UART/I²C/SPI)
 MPU6050 — acceleration, gyroscope (I²C)
 BNO085 — orientation, magnetometer (UART/I²C/SPI)
 M5Stack Unit Cam- OV2640 (UART/WiFi)
 Voltage monitor — system power health (ADC)



Simulation Mode Software

TestAndSim suite - overview

A compile-time swappable module (activated via `config.h` flag) that replaces the Hardware Controller Layer with procedurally generated sensor data. All upper layers i.e FSW Core, Data Layer and CommLayer; execute identically to flight firmware. LoRa is replaced by a loopback stub for packet inspection without radio hardware. The actual data generated is dependent on the current preset scenarios of the mission we will create, and will be saved as a collection of `scenarios` such as Nominal Flight, Apogee Edge Case, Launch Edge case and others.

SimDriver_t controller struct

Owens: simulated sensor readings per channel, active scenario ID, simulation tick counter, FaultFlags_e for fault injection. Tick rate matches the highest active sensor acquisition rate to exercise full FreeRTOS load.

Noise model

Per-channel noise consistent with published sensor datasheets. Fixed seed for deterministic regression runs; randomised seed for stress testing. Both modes selectable via SimDriver_t config field.



Test Methodology

The TestAndSim suite also will contain an extensive testing suite that uses both white-box and black-box testing to ensure the robustness of the software.

- Two-level testing: unit tests (white-box) + full pipeline tests (integration, black-box) via TestAndSim; hardware testing covered separately
- White-box: checks individual functions (FSM logic, sensor fusion, packet formatting, SD card handling) in isolation, including edge cases
- Black-box: runs full simulated flight scenarios and checks telemetry output and state transitions against expected behavior
- Some checks: correct state sequencing, packet counter continuity, threshold/boundary behavior, fault handling without system stalls, recovery after reset.
- All thresholds are pulled from a central config file, so tests auto-update when values are tuned
- Any failed test blocks are caught during development so issues are caught early, before they compound.

Prototyping & prototyping environments

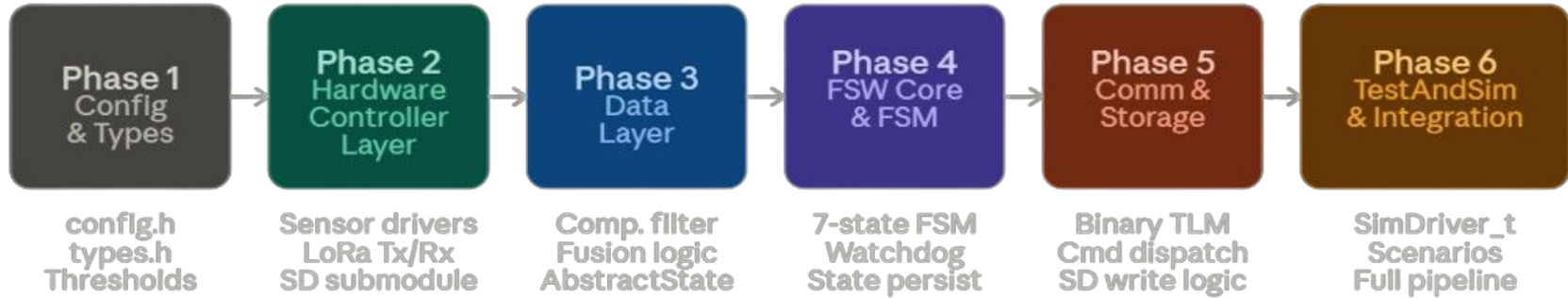
- Development host (unit testing): White-box unit tests run on the development PC without FreeRTOS or physical hardware. Enables rapid iteration during early firmware development.
- TestAndSim environment (integration): Full FSW pipeline on STM32 with TestAndSim replacing Hardware Controller Layer. LoRa replaced by loopback stub.
- Proto hardware bench: Assembled proto unit with all sensors soldered and bus-connected. Sensor bring-up per sensor type

Test Methodology

- All tests versioned alongside firmware source. Results must be stable across multiple runs with differing noise seeds
- Sensors are tested independently and are then integrated with FSW

Software Development Plan(2/2)

Software subsystem development sequence



Key development dependencies & milestones

- Phase 1 (config.h / types.h) is the single source of truth and must be locked before Phase 5 binary serialisation
- Phase 2 sensor drivers are independently testable on bench; bring up one sensor per session
- Phase 3 complementary filter validated against synthetic inputs (unit test) before integrating with real sensors
- Phase 4 FSM gate: all 7 states must traverse correctly in TestAndSim nominal scenario before Phase 6 begins
- Phase 6 regression baseline (nominal flight scenario) becomes mandatory CI gate — failure blocks further testing
- Hardware proto validation (drop test, LoRa range, watchdog timeout) run in parallel with Phase 6 integration



Ground Control System (GCS) Design



GCS Hardware and Software Architecture

GCS Hardware Configuration

Laptop Computer Primary processing, display, and logging platform
 SX1276-compatible LoRa receiver RF reception and telecommand uplink transmission
USB-to-UART interface Bridge between LoRa receiver and laptop software
Fiberglass collinear antenna 865–867 MHz, 5–8 dBi gain, vertical polarisation, tripod-mounted
Portable field power source Independent operation — no mains electricity required

Configured: 865–867 MHz | SF7 | BW 125 kHz | CR 4/5 | Sync 0x12

GCS Software Stack

Python — Core application language Cross-platform; serial I/O, data processing, GUI integration
PyQt6 — GUI framework Event-driven; multi-panel layouts, custom widgets
PyQtGraph — Real-time plotting OpenGL-accelerated; updates at 1 Hz telemetry rate
Pandas — Data processing CSV logging; time-series operations; structured data
NumPy — Numerical computation Vertical velocity derivation; vectorised operations

Integrates real-time serial handling, multi-threaded processing, and multi-panel visualisation within a single application.



GCS Mission Status Panel and Command Interface

Mission Status Panel

Persistent panel updating at 1 Hz with each incoming packet:

- Mission Elapsed Time — from GPS UTC in telemetry
- Packet Count — total received; gaps from sequence counter
- Mission State — reflects current CanSat FSM state
- RSSI (dBm) — signal strength from LoRa receiver
- SNR (dB) — signal-to-noise ratio
- Battery Voltage — real-time power health
- Reset Count — processor resets since mission start
- Communication Link Status — active / lost / degraded

Command Interface

XOR checksum auto-computed by GCS software — every packet structurally valid. ACK/NACK displayed per command.

0x01 CMD_START_TELEMETRY — Activate downlink

0x02 CMD_STOP_TELEMETRY — Suspend downlink

0x03 CMD_CALIBRATE_BARO — Barometric zero

0x04 CMD_CALIBRATE_GYRO — Gyro calibration

0x05 CMD_CALIBRATE_ACCEL — Accel calibration

0x06 CMD_ENTER_TEST — Transition to TEST_MODE

0x07 CMD_ENTER_LAUNCHPAD — To LAUNCH_PAD

0x08 CMD_SWITCH_CHANNEL — Switch radio channel

GCS Real-Time Displays and Attitude Visualisation

Real-Time Telemetry Display

Structured display by category, updating at 1 Hz:

Environmental

- Altitude (m AGL), Pressure (hPa), Temperature (°C)

IMU

- Acceleration X, Y, Z (m/s^2)
- Angular velocity X, Y, Z (deg/s)

GPS

- Latitude, Longitude (decimal degrees)
- GPS Altitude (m), Satellite count

Power and Communication

- Battery voltage (V)
- RSSI (dBm), SNR (dB), Packet count, Link status

Attitude Visualisation Panel

Dedicated panel: artificial horizon, roll, pitch, heading — driven by BNO086 orientation data.
Continuous visual monitoring without manual interpretation of raw angular rates.

Real-Time Plots (PyQtGraph, 1 Hz)

Flight Data

- Altitude vs. Time
- Pressure vs. Time
- Temperature vs. Time
- Vertical Velocity vs. Time (derived)

IMU

- Acceleration X, Y, Z vs. Time
- Angular Velocity X, Y, Z vs. Time

Orientation

- Roll, Pitch, Yaw vs. Time

GPS and Power

- Latitude, Longitude, GPS Altitude, Satellite Count
- Battery Voltage vs. Time



GCS Logging, Recovery Interface and Field Deployment

Data Logging

Timestamped CSV files on local storage. Logged per packet:

- Mission Time (ms since launch epoch)
- Packet Number
- All sensor fields: altitude, pressure, temp, accel X/Y/Z, ang-vel X/Y/Z, roll, pitch, yaw
- GPS: lat, lon, altitude, satellite count
- Mission State (FSM identifier)
- RSSI (dBm), SNR (dB)

GCS Mission Continuity

On GCS software restart, the application recovers mission time, packet count, and flight state from the last CSV log before resuming — maintaining GCS coherence after a laptop interruption.

Recovery Interface

After CanSat transitions to IMPACT state:

- Launch location — fixed reference
- Current GPS position — last received
- Landing position — last valid fix
- Recovery guidance — direction and distance

Field Deployment

Designed for rapid deployment by a small team:

- Position laptop at operations area
- Connect LoRa receiver via USB-to-UART
- Erect tripod, mount collinear antenna
- Connect portable power source
- Launch GCS software; verify receiver configuration
- Issue CMD_START_TELEMETRY once CanSat powered

Design Justification

Python/PyQt6/PyQtGraph: cross-platform, mature libraries, single application integrating serial handling, processing, and visualisation.

Automatic XOR checksum eliminates operator errors in packet construction.
ACK/NACK provides unambiguous command confirmation.

5–8 dBi collinear antenna is the primary ground-side contributor to the 56 dB link margin from the CDH link budget.



Telemetry and Telecommand Packet Structure

Telemetry Packet — 75 Bytes (Finalised)

0	team_number	uint16_t	2B
2	mission_time	uint32_t	4B
6	packet_count	uint16_t	2B
8	altitude	float	4B
12	pressure	float	4B
16	temperature	float	4B
20	voltage	float	4B
24	gnss_time	uint32_t	4B
28	latitude	float	4B
32	longitude	float	4B
36	gnss_altitude	float	4B
40	gnss_sats	uint8_t	1B
41	accel_x/y/z	float x3	12B
53	roll, pitch	float x2	8B
61	gyro_x/y/z	float x3	12B
73	flight_state	uint8_t	1B
74	checksum	uint8_t	1B

Earlier revisions referenced a 73-byte packet; the structure above (75 bytes) is the finalised PDR definition and supersedes that figure throughout this report.

Telecommand Packet — 6 Bytes

Byte 0: Sync Byte = 0xAA

Bytes 1–2: Team Number (uint16_t)

Byte 3: Opcode

Byte 4: Parameter

Byte 5: XOR Checksum (Bytes 0–4)

Key Design Points

- Packed struct — no compiler padding; identical layout in firmware and GCS parser
- Little-endian byte order (STM32F405 native)
- packet_count (offset 6) enables GCS gap detection
- checksum (offset 74) is the application-layer XOR — Layer 4 of validation architecture
- flight_state (offset 73) directly drives GCS Mission State display
- 6-byte command format minimises uplink air time while retaining sync, source ID, and integrity check



Communication Validation Architecture — 4-Layer Defence

Why Layered Validation?

Each layer addresses a distinct failure mode. No single layer is sufficient in isolation — removing any one leaves the system vulnerable to the failure mode it specifically addresses.

Telecommand-Specific Gates

Telecommand packets pass through two additional checks beyond the four layers:

- Sync byte validation — Byte 0 must equal 0xAA
- FSM-state compatibility — command must be valid in current state

Only packets satisfying all applicable conditions are accepted for processing. A command failing any check returns NACK; a valid command compatible with the current FSM state returns ACK.

Outcome

This architecture defends simultaneously against: random channel noise, RF interference from co-located teams, packet collisions, and accidental or malicious cross-team commands — each addressed by a different layer.

4-Layer Validation Pipeline

Layer 1 — LoRa PHY Synchronisation

Applied to: all received frames. Addresses: bit errors from channel noise; frame misalignment

Layer 2 — Team Number Filtering

Applied to: all received packets. Addresses: cross-team command interference at shared launch sites

Layer 3 — CRC Validation

Applied to: all received packets. Addresses: residual bit errors not caught by PHY sync

Layer 4 — XOR Checksum Verification

Applied to: telemetry + telecommand. Addresses: application-layer corruption; end-to-end integrity

Telecommand-Only: Sync Byte (0xAA) + FSM-State Check

Final gates before command execution



Boot Sequence — 9-Step Initialisation Workflow (Finalised)

Boot Sequence — Finalised

The flight software boot sequence is finalised. Each step completes successfully before the next begins; a failure is recorded in the health diagnostic log and reported to the GCS once telemetry is established.

1. Power Application

Main power rail established; regulators stabilise supply

2. STM32F405 Initialisation

Flight computer core, clocks, internal peripherals operational

3. Sensor Initialisation

BMP585, BNO086, NEO-M9N out of reset; registers configured

4. Communication Bus Init

SPI1, SPI2, I²C, UART1 initialised at configured rates

5. microSD Mounting

FAT32 mounted on SPI1; log files opened; write access verified

6. Telemetry Subsystem Init

SX1276 configured; TX/RX buffers initialised

7. Health Diagnostics

All subsystem self-tests executed; faults flagged

8. Previous Mission-State Recovery

Persisted time, packet count, FSM state read from microSD

9. Transition to BOOT Completion

BOOT → TEST_MODE → LAUNCH_PAD; awaits CMD_START_TELEMETRY

Boot-Time Measurement — PDR Status

A measured boot duration has not yet been established, as integrated hardware testing has not been completed. The sequence itself (left) is fully finalised — only the timing measurement is pending.

Boot-time measurement will be finalised during integrated system testing and flight qualification activities, with the measured value reported at the Critical Design Review.

Step 8 — Previous Mission-State Recovery

Entry point to the reset recovery workflow detailed on the next slide. Ensures the system resumes with the correct mission time, packet counter, and FSM state after any processor reset, during initial boot or mid-mission.

Step 9 — Transition to BOOT Completion

The FSM completes BOOT, transitions through TEST_MODE on successful validation, then to LAUNCH_PAD. The system remains in LAUNCH_PAD — sensors active, telemetry link established, downlink not yet transmitting — until CMD_START_TELEMETRY is received from the GCS.



Mission Continuity and Reset Recovery

Persisted Mission Variables

Three variables are continuously written to microSD during operation, enabling coherent recovery after an unplanned processor reset:

Mission Elapsed Time

Written each telemetry cycle — provides time continuity after reset

Packet Counter

Written each telemetry cycle — GCS never sees a counter reset to zero

Current FSM State

Written on every state transition — correct operational state resumed

Why This Matters

Without persisted state, a processor reset mid-mission would restart the packet counter from zero, reset mission time to zero, and force the FSM back to BOOT — producing a telemetry record the GCS cannot reconcile with pre-reset data. Persisted state ensures the post-reset stream is a continuation, not a restart.

The GCS distinguishes a reset event from packet loss via the Reset Count field: a reset produces a small time gap with continuity preserved, whereas packet loss produces sequence-counter gaps with no time discontinuity.

Reset Recovery Workflow

1. Read Stored State

Mission state log file read from microSD

2. Restore Packet Counter

Resume from last persisted packet_count value

3. Restore Mission Time

Resume from last persisted mission_time value

4. Restore FSM State

Resume in the last persisted flight_state

5. Resume Nominal Operation

Continue sensor sampling, telemetry, and logging from restored context

This workflow executes as Step 7 of the boot sequence and applies identically whether the reset occurs during pre-launch testing or mid-flight.



System Design Justification — Consolidated Engineering Rationale

SX1276 LoRa Transceiver

Long-range, low-power, high interference immunity — well-suited to shared launch sites with multiple concurrent RF systems

SF7 Spreading Factor

Highest LoRa throughput; 56 dB link budget margin makes additional margin from higher SF unnecessary

Dual-Path Architecture

Simultaneous LoRa telemetry + microSD logging — mission data survives any link failure; no in-flight recovery possible otherwise

Configurable FSM Thresholds

Thresholds beyond the mission-specified 600 m AGL are deferred — separates tuning from firmware recompilation, enabling rapid post-test iteration

Layered Validation Architecture

4 independent layers (PHY sync, team ID, CRC, XOR) each address a distinct failure mode — no single point of validation failure

Vertical Antenna Polarisation

Matched polarisation on both ends eliminates 3–6 dB of avoidable mismatch loss

Omnidirectional Antenna Pattern

CanSat rotates continuously during descent — omnidirectional pattern avoids orientation-dependent signal nulls a directional antenna would introduce

Higher-Gain GCS Antenna

5–8 dBi collinear compensates for CanSat's constrained monopole and boosts link margin without CanSat-side complexity or tracking hardware



Preliminary Link Budget Analysis

Assumptions

TX Power (Pt) = 17 dBm

CanSat Antenna Gain (Gt) = 2 dBi

GCS Antenna Gain (Gr) = 5 dBi

Link Distance (d) = 1 km | Frequency (f) = 867 MHz

SX1276 Sensitivity (SF7, 125 kHz BW) = -123 dBm

Equations

$$FSPL(dB) = 20 \cdot \log_{10}(d) + 20 \cdot \log_{10}(f) + 32.44$$

$$Pr = Pt + Gt + Gr - FSPL$$

$$LM = Pr - \text{Sensitivity}$$

Calculations

$$FSPL = 20 \cdot \log(1) + 20 \cdot \log(867) + 32.44 \approx 91 \text{ dB}$$

$$Pr = 17 + 2 + 5 - 91 = -67 \text{ dBm}$$

$$LM = -67 - (-123) = 56 \text{ dB}$$

Results and Interpretation

FSPL @ 1 km, 867 MHz
≈ 91 dB

Received Power (Pr)
≈ -67 dBm

Link Margin (LM)
56 dB — ROBUST

Significance of 56 dB Margin

The received signal is 56 dB above the SX1276 demodulation floor. Each 6 dB of margin doubles the tolerable signal degradation, so 56 dB represents a very large safety factor.

This margin accommodates:

- Antenna orientation losses: 10–15 dB for a misaligned monopole — fully covered
- Typical multipath fading: 10–20 dB — fully covered
- Launch-site RF interference from co-located teams — covered

SF7 is confirmed appropriate for the 1 km LOS mission profile, with margin to adjust to a higher spreading factor if ground testing reveals degradation not captured by the free-space model.



Mission Timing Architecture — GNSS-Synchronised

No Dedicated RTC Module

The CDH subsystem does not use a dedicated external real-time clock (RTC) IC. Instead, mission timing is implemented as a GNSS-synchronised architecture.

Time Architecture

- GNSS UTC time (NEO-M9N) is the primary absolute time reference
- STM32F405 generates mission elapsed time relative to launch epoch
- Mission time included in every telemetry packet (offset 2)
- Mission time persisted to microSD and restored on reset recovery
- Raw GNSS time (offset 24) also logged for post-flight correlation

Why GNSS-Synchronised vs. RTC?

A GNSS-derived absolute reference is referenced to a global atomic time standard, eliminating the drift associated with onboard crystal oscillators over multi-hour missions — without the added component, power, and battery-backup overhead of a separate RTC IC.

Timing Data Flow

NEO-M9N GNSS Receiver

Provides UTC time — primary absolute reference

STM32F405 Flight Computer

Generates mission elapsed time from launch epoch

Telemetry Packet (offset 2)

mission_time included in every 75-byte packet

microSD Persistence

mission_time saved each cycle — reset recovery source

Telemetry Packet (offset 24)

Raw gnss_time also logged for post-flight correlation

Dual recording of both GNSS time and mission elapsed time in every packet allows post-flight analysis to cross-reference onboard event timing against an independent absolute clock.



Memory Architecture — Function-Oriented Design

Function-Oriented, Not Slot-Based

The CDH memory architecture is organised by function rather than predefined storage "slots." Three distinct resources serve complementary roles spanning firmware storage, runtime operation, and mission data persistence.

STM32F405 Internal Flash

Non-volatile (onboard). Flight software storage, configuration parameters, configurable FSM thresholds, system initialisation data

STM32F405 Internal SRAM

Volatile (onboard). Sensor acquisition buffers, telemetry packet assembly, temporary mission data processing, communication buffers

microSD Card (FAT32)

Non-volatile (removable). Telemetry logging, mission-state logging, recovery information storage, post-flight data retrieval

A subset of microSD data — mission elapsed time, packet counter, and FSM state — is periodically persisted to support reset recovery (see Mission Continuity slide).

Compliance Statement

Memory Requirements — SATISFIED

Satisfied through the combination of internal Flash (firmware + config), internal SRAM (runtime operation), and FAT32 microSD (mission data storage and persistence) — no fixed number of discrete storage slots required

Why Function-Oriented?

Defining memory by role rather than by a fixed slot count reflects how embedded flight software actually allocates resources: firmware and configuration are compiled into Flash at build time, runtime data structures occupy SRAM during operation, and mission records accumulate on removable storage independent of the processor's internal memory map.

This framing avoids an arbitrary "number of slots" specification that would not map cleanly onto the STM32F405's actual memory architecture, while still demonstrating that every functional memory requirement — firmware, configuration, runtime buffers, and mission data — is met by a specific, identified resource.



Transmission Loss Mitigation and Backup RF Capability

Dual-Path Telemetry Handling

Primary — Real-Time Telemetry

SX1276 LoRa downlink to GCS at 1 Hz — live mission monitoring

Secondary — Onboard Data Preservation

Simultaneous logging of the same packet to FAT32 microSD, independent of RF link status

Key Outcome

Every telemetry packet generated by the flight computer is both transmitted via LoRa and logged to microSD. Because both paths execute for every packet, loss of the RF communication link does not result in loss of mission data — all mission telemetry remains available for post-flight recovery and analysis via the onboard storage subsystem.

PDR-Stage Mitigation Approach

Transmission-loss mitigation at PDR stage is achieved through dual-path telemetry handling — real-time LoRa transmission combined with simultaneous onboard logging — rather than through a currently implemented secondary RF link.

Backup RF Channel Capability

CMD_SWITCH_CHANNEL (Opcode 0x08)

Reserved capability for alternate RF channel selection — implemented and validated through the standard telecommand pipeline

Deferred to CDR — Reasons

- Final spectrum planning has not yet been completed
- Channel allocation depends on integrated RF testing
- Current PDR focuses on the primary communication architecture

Backup During Transmission Loss — SATISFIED

Via simultaneous onboard logging + mission-state persistence. Backup RF channel capability exists in command architecture; channel definitions remain a CDR item

This separates what is implemented now (dual-path data protection, reserved opcode) from what is intentionally deferred (specific channel frequencies/parameters).



Bill of materials

Cost for all electrical components

Component	Qty	Unit Cost (INR)	Total (INR)
STM32F405 Development Board	1	850	850
BMP585 Sensor Module	1	690	690
BNO086 IMU Module	1	2100	2,100
NEO-M9N GPS Module	1	2350	2,350
SX1276 LoRa Module	1	420	420
M5Stack Unit CAM (OV2640)	1	1150	1,150
microSD SPI Module	1	120	120
MP1584 Buck Converter Module	1	95	95



TPS73733 LDO Regulator	1	180	180
SMBJ12A TVS Diode	1	30	30
XT60 Connector Pair	1	80	80
3A Automotive Fuse	1	25	25
Samsung INR18650-30Q Cells	2	625	1,250
Battery Holder + Nickel Strip	1	220	220
External Toggle Switch	1	60	60
Power LED + Resistor	1	10	10
Capacitors + Ferrites + Passives+Wiring + Connectors + Mounts	—	—	180+450
SUBTOTAL+Contingency(10%)			INR 10,300+INR 1,210
			11,510